Journal of Business Research 195 (2025) 115385



Contents lists available at ScienceDirect

Journal of Business Research

journal homepage: www.elsevier.com/locate/jbusres





How categories grow: The behavioural drivers of revenue growth

Steven Dunn [a,*], Magda Nenycz-Thiel [a], Charles Graham [a,b], John Dawes [a], Nick Danenberg [a], Arry Tanusondjaja [a], Giang Trinh [a], Bruce McColl [a]

[a] Ehrenberg-Bass Institute, University of South Australia, GPO Box 2471, Adelaide, SA, Australia
[b] London South Bank University, 103 Borough Road, London SE1 0AA, United Kingdom

ARTICLE INFO

Keywords:
Category growth
Buyer behaviour
Household panel data
Penetration
Premiumisation

ABSTRACT

This study examined category revenue growth and decline in the consumer packaged goods (CPG) industry from a buyer behaviour perspective. Using 13 years of US household panel data for 474 categories, the research documents the incidence and behavioural drivers of category revenue changes. Unlike previous work that focused on total sales, this study disaggregates revenue into four buyer behaviour components: penetration, purchase frequency, volume per trip, and price per volume. The findings identified penetration and price per volume as the primary drivers of growth, while category size was found to influence their relative contributions and expected magnitude of growth. Penetration loss consistently emerged as the leading cause of decline across all category sizes. The research contributes to marketing theory by highlighting the importance of buyer behaviour in category dynamics. It also provides a novel managerial framework for more effective category growth strategies.

1. Introduction

Category growth is increasingly prioritised by leading firms in the consumer packaged goods (CPG) industry. Mark Pritchard, P&G's Chief Brand Officer, described it as "*the highest order ambition*" (Innes, 2023), a perspective shared by Mars Wrigley (Tesseras, 2023) and PepsiCo Europe (Gollasch, 2024). While such firms could continue competing for market share in their established markets, they are increasingly shifting their focus towards category expansion. This shift broadens marketing's remit to encompass both primary and secondary demand.

Category growth, defined as the annual increase in category revenue, can occur when marketers increase the number of category buyers or encourage existing buyers to purchase more often, buy larger quantities, or pay higher prices. However, the relative effectiveness of these strategies in driving category sales—or preventing *category decline*, defined as a decrease in annual category revenue—remains speculative (e.g., Bang et al., 2016; Kotler et al., 2008, p.475; Varadarajan et al., 1992). Despite the emphasis on category growth in the CPG industry, the scale and frequency of these opportunities remain unclear. Reliable knowledge to support marketers in achieving growth is limited, which motivates this study to establish robust evidence of the incidence and drivers of category growth under varying conditions.

Understanding how to respond to or influence category growth, whether driven by brand management or external factors, is important for brands of all sizes. Category growth is particularly critical for large, market-leading brands, as increases in category sales contribute more significantly to their revenue growth than gains in market share (Tanusondjaja et al., 2022). Smaller brands also benefit because simply maintaining brand share during periods of expansion leads to sales growth.

Empirical studies on CPG category growth have predominantly focused on developing econometric models to distinguish between stationary and evolving sales trends in time-series data (Dekimpe & Hanssens, 1995a; Srinivasan & Bass, 2000). While valuable for assessing the persistence of marketing effects on category sales (e.g., Cavaliere & Tassinari, 2001; Dekimpe et al., 1998; Nijs et al., 2001; Wilcox et al., 2015), these studies offer little insight into the incidence or magnitude of revenue changes. Recognising the need for greater certainty regarding the outcomes of marketing investments aimed at category expansion, this study's first research question examines the incidence and scale of annual category revenue changes (RQ1).

Household panel data, widely used by CPG firms, tracks the

* Corresponding author.
E-mail addresses: steven.dunn@marketingscience.info (S. Dunn), magda.nenycz-thiel@marketingscience.info (M. Nenycz-Thiel), charles.graham@marketingscience.info (C. Graham), john.dawes@marketingscience.info (J. Dawes), nick.danenberg@unisa.edu.au (N. Danenberg), arry.tanusondjaja@marketingscience.info (A. Tanusondjaja), giang.trinh@unisa.edu.au (G. Trinh).

https://doi.org/10.1016/j.jbusres.2025.115385
Received 6 December 2023; Received in revised form 21 February 2025; Accepted 6 April 2025
Available online 22 April 2025
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purchasing behaviour of individual households over time, offering insights into who buys, how often, how much, and at what price. While early studies used these data to model category revenue trends (e.g., Dekimpe et al., 1998; Srinivasan & Bass, 2000), few have explored the behavioural drivers of these changes. By contrast, brand-level studies have demonstrated the value of analysing household panel data in developing laws and models of buying behaviours linked to brand growth (Baldinger et al., 2002; Romaniuk et al., 2014, 2018). Importantly, household panel data enable the measurement of the four behavioural components of category growth: category penetration (number of buyers), purchase frequency (how often buyers purchase), volume per trip (quantities purchased), and price per volume (price paid). Leveraging the ability to disaggregate total category revenue, this study's second research question investigates consistent relationships between category growth and changes in these metrics (RQ2), while the third research question examines potential interdependencies among them (RQ3).

This study adopts the ETET (Empirical → Theory) research approach, as advocated by Ehrenberg (1994, 1995). The aim is to build theory by identifying regularities and relationships replicated across many sets of data (MSoD) (Ehrenberg, 1990), rather than relying on a single dataset. The major advantage of this approach lies in generating 'reusable knowledge by design' (Sharp et al., 2017; Uncles & Kwok, 2013), ensuring that findings have both predictive validity and immediate managerial relevance (Ehrenberg, 1988). Within a single study, collecting and reporting intra-study replications accelerates theoretical progress by establishing reliability and identifying boundary conditions at the outset (Ehrenberg, 1995; Uncles & Kwok, 2013; Wright & Kearns, 1998).

This approach was applied through an analysis of US household panel data spanning 13 years, covering 474 established categories and more than 5,500 annual category revenue data points. A novel and simple method of category performance analysis is introduced that extends the techniques commonly applied in brand-level buyer behaviour studies (Romaniuk et al., 2014, 2018). The analysis first establishes the magnitude of all annual category revenue changes, and then disaggregates these changes to identify regularities across the four underlying components of buyer behaviour.

The findings reveal systematic variations in revenue changes based on category size. Low-penetration categories exhibit greater sales fluctuations, whereas high-penetration categories tend to be more stable. Further analysis of buyer behaviour revealed consistent growth patterns linked to category size. For the low-penetration categories, increases in revenue were closely related to changes in penetration. In contrast, large-penetration categories (with $\geq$ 40 % quarterly penetration) primarily grow through increases in price per volume, without reductions in penetration. Changes in purchase frequency and volume per trip had a minimal impact on overall revenue, suggesting that these are not significant drivers of category growth. Regardless of the category size, penetration loss consistently emerged as the primary driver of category revenue decline. Observed across MSoD within this single study, these findings reflect the characteristics of behavioural norms. A key contribution of the ETET approach is the proposed evidence-based framework seeking to inform future research agendas and category growth strategies.

2. Background

Sales outcomes from brand marketing interventions depend on both category expansion (primary demand) and market-share response (secondary demand) (Hanssens, 2014, p.102). However, primary demand is often viewed as merely the backdrop on which brand marketing strategies are built—a 'given' (Jaworski & Kohli, 2000) based on which brand shares are calculated, resources allocated, and competitor analysis undertaken (Day et al., 1979). This 'myopia' may stem from several factors, one being the short-term focus of management (Lodish & Mela, 2007). Marketers can overlook category-level dynamics because category buying remains largely stable within short timeframes (Goodhardt et al., 1984). Additionally, the challenges of accessing long-term data and using complex modelling to analyse category-level sales trends (Dekimpe & Hanssens, 2000), along with the perception that competitive advantages in category sales responses are dictated by market share (Tanusondjaja et al., 2022), likely reinforce this perspective.

Therefore, category growth has been significantly under-researched compared to market share gains despite being critical to overall sales outcomes. A review of existing literature was conducted to contextualise the research questions explored in this study.

2.1. Empirical studies of category growth and decline

Early research on CPG category sales growth and decline relied predominantly on econometric time-series analyses of retail and household panel data (e.g., Dekimpe & Hanssens, 1995a; Dekimpe et al., 1998; Nijs et al., 2001; Pauwels et al., 2002; Srinivasan & Bass, 2000). These studies typically applied methods such as unit root tests to distinguish between stationarity and evolution in sales. Stationarity refers to sales changes that fluctuate but revert to a stable level or trend, whereas evolution indicates a persistent change from a long-run mean, either as growth or decline. Such classifications have been instrumental in assessing the persistence and impact of marketing interventions on category demand (Dekimpe & Hanssens, 1995b). By contrast, descriptive methods, although less frequently employed, offer insights into the magnitude of category sales changes using similar datasets (Dunn et al., 2021; Tanusondjaja et al., 2022). Table 1 summarises the category growth literature underpinning this study.

The first sub-strand investigated the relationships between category sales dynamics, market share, and brand performance. Dekimpe and Hanssens (1995a) conducted a *meta*-analysis of univariate time-series models from 1975 to 1994, revealing that category sales models exhibited evolution in 92 % of the cases, whereas only 22 % of brand market share models showed similar patterns. This finding suggests that permanent shifts in brand sales are driven more often by changes in overall category sales than by changes in market share. Srinivasan and Bass (2000) extended this finding by analysing eight CPG categories in the US, reporting that while market shares remained largely stationary, category sales evolved in approximately half of the cases. Thus, even when brands maintain a steady market share, they experience sales growth if category sales expand. Tanusondjaja et al. (2022) examined the relative contributions of category expansion and share gain to revenue growth across 37 categories, concluding that share gains drive revenue growth for smaller firms, whereas market-leading firms derive greater benefits from category expansion.

The second sub-strand focused on the effects of price promotions and advertising on category growth. Although price promotions are a common feature of CPG marketing (Bogomolova et al., 2015), they generally have little lasting impact at the category level. Dekimpe et al. (1998) found no long-term impact of price promotions on category sales in three cases, and only minor category sales effects in a fourth. In a more comprehensive study, Nijs et al. (2001) analysed 560 categories over four years and found positive long-term sales effects from promotions in only 2 % of cases. Their large sample also offered a more representative view of the incidence of category growth and decline, showing that most categories (506 out of 560) were stationary; however, many of these exhibited deterministic sales trends that were either positive (36 %) or negative (26 %).

Research on advertising expenditure has yielded limited evidence of its impact on long-term category growth. Cavaliere and Tassinari (2001) found no significant effect of advertising on whiskey sales over a seven-year period, attributing observed category sales changes to price fluctuations. Similarly, Wilcox et al. (2015) found no direct relationship between advertising expenditure and category growth in an analysis of beer, wine, and spirit sales over 40 years. They concluded that

Table 1
Empirical studies of category growth and decline.

Paper	Data Type	Data Details	Category Sales	Category Buyer Behaviour	Research Focus
Dekimpe & Hanssens, 1995a	Meta-Analysis	Univariate time-series models (1975–1994). 49 category/industry models	Revenue Sales or Unit Sales	—	Identify stationarity and evolution in time-series models.
Dekimpe et al., 1998	Household Panel Data	4 CPG categories over 113 weeks (United States)	Volume Sales	—	Long-run effects of price promotions on brand and category sales
Srinivasan & Bass, 2000	Household Panel Data	8 CPG Categories over 104 weeks (United States)	Unit Sales	—	Identify stationarity and evolution in brand and category sales.
Nijs et al., 2001	Retail Panel Data	560 CPG Categories for 4 Years (Netherlands)	Volume Sales	—	Short and long-run effects of price promotions on category demand
Cavaliere & Tassinari, 2001	Retail Panel Data	Whiskey from January 1983 to December 1989 (Italy)	Unit Sales	—	Long-term relationships between advertising and category sales
Pauwels et al., 2002	Household Panel Data	Soup and Yogurt from July 1986 to Sept 1988 (United States)	—	Category Incidence (Penetration)	Long-run effects of price promotions on aggregate brand sales components (category incidence, brand choice and purchase quantity).
Wilcox et al., 2015	Per Capita Sales Data	Beer, Wine and Liquor from 1971 to 2012 (United States)	Volume Sales	—	Advertising effects on alcohol category sales.
Dunn et al., 2021	Household Panel Data	50 CPG Categories over 10 years (United States)	—	Penetration, Purchase Frequency	Documenting changes in aggregate buyer behaviour metrics over ten years.
Tanusondjaja et al., 2022	Retail Panel Data	37 CPG Categories over 3–5 Years (United States & United Kingdom)	Revenue Sales	—	Relative contribution of category growth and market share gains towards firm growth.
This Study	**Household Panel Data**	**474 CPG Categories (United States)**	**Revenue Sales**	**Penetration, Purchase Frequency, Volume per Trip, Price per Volume**	**Document empirical patterns in extent of category growth (revenue sales) and mechanisms underlying category growth (buyer behaviour).**

demographic shifts, taxation, and income levels are the primary drivers of category sales.

Some studies have highlighted the value of examining the underlying components of category revenue changes; however, they have not been systematic. Pauwels et al. (2002) analysed the long-term impact of price promotions on brand sales in two categories. Although their study did not focus on category sales, the disaggregation of brand sales components revealed a notable boost in category-level purchase incidence during the promotion period, which persisted in the subsequent period. When analysing buyer behaviour metrics in long-term panel data, Dunn et al. (2021) found that category penetration figures fluctuated significantly more than purchase frequencies over a decade but did not investigate how these buyer behaviour metrics related to total sales.

Despite valuable early insights, empirical studies on category growth are yet to provide consistent and managerially relevant findings to guide growth objectives or strategies. This stems from the limited number of studies conducted over the past three decades and the inconsistency in research focus (see Table 1). Many studies have explored narrow topics, such as price promotions or advertising, rather than examining broader patterns of category growth. Few studies have systematically investigated category growth across a wide range of categories, typically analysing fewer than 50. These risks overlook the important characteristics of hundreds of CPG categories, including category size and category purchase frequency which influence penetration growth potential (Graham & Kennedy, 2022). Moreover, most studies have relied on older data. More recent household panels now offer far longer timeframes of continuous buying, which are helpful in assessing category buying behaviours.

2.2. Magnitude of category growth and decline

Given the importance of total category sales for firm performance (Liu et al., 2015; Tanusondjaja et al., 2022), management must carefully consider, plan, and set realistic ambitions for growth based on category performance. This requires an understanding of the norms for both the incidence and magnitude of category growth and decline as well as the conditions that drive these variations.

The most extensive study on category dynamics to date (Nijs et al., 2001) examined 560 categories and found that approximately one-third maintained stable sales volumes over a four-year period. While it highlights the frequency of growth and decline, it does not address the magnitude of change, limiting its usefulness in setting realistic managerial goals. In contrast, Tanusondjaja et al. (2022) explored the scale of category fluctuations and reported an average annual sales change of 1.1 %, with variations ranging from −9.1 % to 10.2 %. However, uncertainties persist regarding the generalisability of these findings across broader CPG categories, and how growth and decline patterns may differ under varying conditions or across specific categories.

The literature on brand market share performance offers a useful parallel, in which combining time-series analysis with descriptive approaches has advanced the understanding of share change norms. Econometric time-series methods have long shown that stationarity is the dominant characteristic of brand market share (Bass & Pilon, 1980; Dekimpe & Hanssens, 1995a; Franses et al., 2001; Lal & Padmanabhan, 1995). A clearer picture of the expected magnitude of share changes has since emerged through studies documenting share changes across various brands and categories. For instance, Graham (2009) found that only 25 % of brands experienced share changes greater than 3 percentage points over 6.5 years, whereas Trinh and Anesbury (2015) observed similar shifts in just 14 % of brands over five years.

In contrast to these brand performance studies, few studies have empirically examined the extent to which categories grow or decline annually, a common managerial timeframe, or the conditions that influence this variability. Thus, this study seeks to build on established approaches to analyse brand share changes and apply them to category sales:

RQ1: What is the incidence and magnitude of annual category revenue growth and decline across CPG categories?

2.3. The buyer behaviour components of category growth and decline

Strategic marketing objectives are achieved by influencing consumer behaviour. Theoretical pathways to growth have often been proposed as strategies for category expansion (Bang et al., 2016; Bass et al., 2005; Kotler et al., 2008; Varadarajan et al., 1992). The first involves attracting more non-category buyers to the customer base (category penetration). The second involves increasing buyers' consumption, either through more frequent purchases (category purchase frequency) or purchasing larger quantities (category volume per trip). Finally, category revenue can grow if buyers pay more for the volume they purchase (category price per volume), whether through price increases or a shift toward higher-priced options.

The appropriateness of growth strategies may depend on the stage of category development, with penetration possibly being more critical for newer categories and usage becoming more relevant as a category matures (Bass et al., 2005). Strategies could also be implemented simultaneously across different consumer segments such as penetration for non-buyers, volume expansion for below-average buyers, and upgrading to migrate above-average buyers to higher-end products (Bang et al., 2016). However, these recommendations remain speculative, and support for increased usage is often limited to textbook case studies.

As summarised in Table 1, much of the research on category growth has focused on changes in aggregate sales outcomes, rather than on the underlying buyer behaviour. This limitation is partly due to the nature of the analysed data, as buyer behaviour cannot be directly observed from the retail panel or per capita consumption data. Studies using household panel data have primarily concentrated on aggregate sales trends (e.g., Dekimpe et al., 1998; Srinivasan & Bass, 2000) or have not explicitly linked specific buyer behaviour dynamics to shifts in category sales (Dunn et al., 2021; Pauwels et al., 2002).

For example, Dunn et al. (2021) found that over a decade, changes in category penetration were more than three times greater than changes in purchase frequency, indicating that penetration is likely to be a stronger driver of category sales. However, the analysis did not include volumes purchased or prices paid, thus providing an incomplete view of category revenue dynamics. This gap highlights the need for a more comprehensive understanding of how buyer behaviour collectively influences category growth.

Analogous research at the brand level using household panels highlights the benefits of systematically examining buyer behaviours underlying market share growth and decline. Brand share changes have consistently been associated with larger changes in penetration than in loyalty (Baldinger et al., 2002; Romaniuk et al., 2014, 2018), leading to unambiguous advice for practitioners to prioritise growing brand penetration, which is an insight not evident from brand sales data alone.

Similar evidence-based knowledge is critical for practitioners who seek to understand category growth. A systematic analysis of the various components of category buying behaviour and their interdependencies is needed to reveal the conditions under which specific behaviours are more likely to drive category expansion. These insights can guide the development of effective category growth strategies. Prior studies have failed to identify consistent relationships between overall category growth and advertising or price promotion activities (e.g., Cavaliere & Tassinari, 2001; Dekimpe et al., 1998; Nijs et al., 2001; Pauwels et al., 2002; Wilcox et al., 2015). However, the effectiveness of such investments may depend less on the type or scale and more on their objectives. Specically, whether they are designed to attract new buyers, increase consumption among current buyers, or encourage purchases of higher-priced products.

Rather than focusing on specific marketing mix investments, this study explores the strength and direction of the relationships between category buyer behaviour components—penetration, purchase frequency, volume per trip, and price per volume—and category revenue changes under different conditions. Additionally, it seeks to determine whether these buyer behaviour components consistently co-vary, indicating whether they represent mutually exclusive behaviours. For instance, higher prices per volume may systematically reduce the number of households purchasing the category. Therefore:

RQ2: What is the relationship between category growth/decline and changes in category buyer behaviour components across CPG categories?

RQ3: What are the relationships between changes in each of the category buyer behaviour components across CPG categories?

3. Methods

3.1. Research approach

Given the scarcity of evidence for CPG category growth or the associated underlying behaviours, an ETET (Empirical → Theory) approach is suited to this study. Such an approach aims to identify recurring patterns and associations in behavioural metrics across multiple datasets and summarise them descriptively, providing a basis for further explanation and theory development (Ehrenberg, 1994, 1995). While ETET is a well-established approach in marketing science, its use aligns with recent calls for 'Empirics-First' research as a basis for theoretical advancement (Golder et al., 2023).

This study builds on the methods used by Romaniuk et al. (2014, 2018) in their analyses of brand share growth and the associated behavioural metrics. Their analysis, which focuses on year-on-year changes in market share and brand performance metrics, is adapted here to examine year-on-year changes in category revenue and the underlying behavioural metrics.

3.2. Data and metrics

Category growth was examined using NielsenIQ US household panel data from 2007 to 2019, capturing CPG purchases from over 60,000 active panellists annually across 474 consumer goods categories. The categories, pre-classified as 'product modules' by NielsenIQ, reflect a practitioner's view of markets, but perhaps not always how consumers perceive them.[1]

A total of 474 categories were selected, based on several criteria. First, it focused on established product categories, including only those with recorded purchases across all 13 years and an average quarterly penetration greater than 1 %. Second, categories with products measured with inconsistent volume units (e.g. litres and grams) were excluded to ensure accuracy in the volume-based analyses. Finally, categories with descriptions implying the aggregation of miscellaneous products were removed (e.g. 'Canned Meat-Miscellaneous'). The final sample included categories from the nine grocery departments (Table 2).

Total sales revenue and buyer behaviour metrics were extracted for each category and for each of the 13 years. To account for the year-to-year variation in the number of households in the panel (minimum: 60,502, maximum: 63,346), sales revenue was standardised to a measure per 100 households, as per Dunn et al. (2021):

- **Sales Revenue (per 100)** = (total sales revenue / no. of panel households) × 100.

Category sales were decomposed into four components of category buying behaviour:

[1] The use of panel data, provided by third parties such as NielsenIQ or Kantar has benefits and limitations as is the case with all data sources. While the data are not specifically designed or gathered to answer the present research questions, it provides a large representative sample of consumer households along with detailed information on CPG purchases, which suits the study objectives.

Table 2
Overview of product category sample.

Department	Categories		Category Examples
	n	%	
Dry Grocery	245	52	Potato Chips, Mustard, Wet Cat Food, Bottled Water, Tea Bags
Health and Beauty	60	13	Cough Drops, Razor Blades, Hand Cream, Shampoo, Adhesive Bandages
Non-Food Grocery	53	11	Toilet Tissue, Aluminium Foil, Liquid Soap, Cigarettes, Paper Towels
Frozen Foods	47	10	Frozen Bagels, Frozen Ground Beef, Frozen Pies, Ice Cream, Ice
Dairy	31	7	Grated Cheese, Fresh Eggs, Yogurt, Sour Cream, Dairy Milk
Alcoholic Beverages	20	4	Ale, Vodka, Canadian Whiskey, Gin, Sparkling Wine
Deli	7	1	Salad Dressing, Sauerkraut, Fresh Salad Mix, Horseradish
Packaged Meat	6	1	Bacon, Frankfurters, Breakfast Sausage, Cocktail Franks
Fresh Produce	5	1	Carrots, Lettuce, Strawberries, Mushrooms, Cranberries
Total	**474**	**100**	

- **Penetration (Pen)** – proportion of households that purchased the category
- **Purchase Frequency (PFreq)** – average number of category purchase occasions per buying household
- **Volume per Trip (Vol/Trip)** – average volume (e.g. ounces) per category purchase occasion
- **Price per Volume (Price/Vol)** – average price paid per volume purchased (e.g. $ per ounce)

These four behavioural metrics—standard in household panel analysis—collectively account for the total category revenue among the panellists. Although population growth also influences revenue, addressing it is beyond the scope of this study.

The sales and buyer behaviour components were first calculated for quarterly (13-week) periods. For each year, the total sales revenue is the sum of sales across the four quarters, while the buyer behaviour components are reported as quarterly averages. Year-to-year changes in sales revenue and average quarterly components were analysed for each category, based on average changes, standard deviations, and correlations between these metrics. The annual timeframe provides insights into strategic decisions within a typical management window.

4. Results

4.1. Category performance metrics

Fig. 1 illustrates the variation in sample category size, using average quarterly penetration and purchase frequency. The sample is skewed towards smaller categories, with over a third having a penetration of 5 % or less. In contrast, only 7 % have a penetration of 40 % or more. The larger categories included dairy milk (84 % penetration, 6 purchases per quarter), fresh eggs (79 %, 3.4), ready-to-eat cereal (72 %, 3.5), and toilet tissue (68 %, 2.2).

Fig. 1 also depicts a positive relationship between category penetration and purchase frequency ($r = 0.568$, $p < 0.01$): categories with higher penetration tend to have higher purchase frequencies. Most categories with a $\geq$ 40 % penetration had purchase frequencies $>$ 2 purchases per quarter, whereas categories with a $<$ 5 % penetration typically had frequencies $<$ 2 purchases per quarter.

Notable outliers include categories such as cigarettes (10 % penetration, seven purchases per quarter), wet cat food (17 %, 5.0), and wet dog food (13 %, 4.0), which serve as restricted buyer bases for smokers and pet-owning households. However, this pattern is not universal among restricted buyer bases. For example, dry pet food is purchased less frequently, possibly because buyers can stockpile large quantities.

4.2. Revenue sales changes

The analysis included 12 annual revenue changes for each of the 474 categories, resulting in 5,688 year-on-year observations to address RQ1. Across all categories and periods, the mean revenue change was 0 % and the median was 0.4 % (Table 2), suggesting a similar probability of growth and decline.

Revenue changes exhibited considerable variability, ranging from −79 % to 238 %, though the middle 60 % of observations (20th to 80th percentiles) fall within a narrower range of ± 5 %. This indicates that, while extreme fluctuations occur, most categories experience relatively modest year-on-year changes. Table 3 further illustrates how aggregate category revenue fluctuates over time, with the largest average decline (−3.2 %) recorded between 2009 and 2010 during the global financial crisis. These patterns suggest shared trends across categories, likely influenced by external economic factors beyond marketers' control.

Given the sample's skew towards smaller categories, Table 4 reports the revenue changes by category size (quarterly penetration), revealing two key patterns. First, the high-penetration categories are slightly more likely to experience revenue growth. Second, the smaller categories exhibit greater year-to-year revenue volatility. For example, in categories with $<$ 5 % penetration, the top quintile showed growth exceeding 5.9 %, whereas for categories with $>$ 40 % penetration, a growth above 3.9 % was sufficient to rank in the top quintile. These findings underscore the importance of benchmarking growth targets relative to category size.

Overall, in response to RQ1, revenue growth and decline vary with category size, as measured by average quarterly penetration. Low-penetration categories exhibit greater revenue volatility, with higher relative rates of both growth and decline, whereas high-penetration categories are more stable and exhibit growth more frequently than decline.

4.3. Relationships between changes in revenue sales and category components

To address RQ2, each category's revenue sales were disaggregated into four components: penetration (Pen), purchase frequency (PFreq), volume per trip (Vol/Trip), and price per volume (Price/Vol). Table 5 summarises the means and standard deviations of the annual changes in these components, along with the correlations between each component and revenue sales.

Despite the overall average static revenue sales, the disaggregated components exhibit divergent trends. Penetration showed a negative shift (−1.8 %), while price per volume increased (2.1 %), with purchase frequency and volume per trip remaining near zero.

Total revenue sales are determined by the product of penetration, purchase frequency, volume per trip, and price per volume, indicating that changes in any component have an equivalent impact on revenue. The typical variability of each component must be examined to evaluate its contribution to revenue changes. Table 5 shows that penetration exhibits the highest year-to-year variability (SD: 8.7 %), whereas purchase frequency is the most stable (SD: 2.0 %). Component variability also differs by category size; in low-penetration categories ($<$20 %), penetration fluctuates the most, whereas in high-penetration categories ($>$20 %), price per volume exhibits the greatest variation.

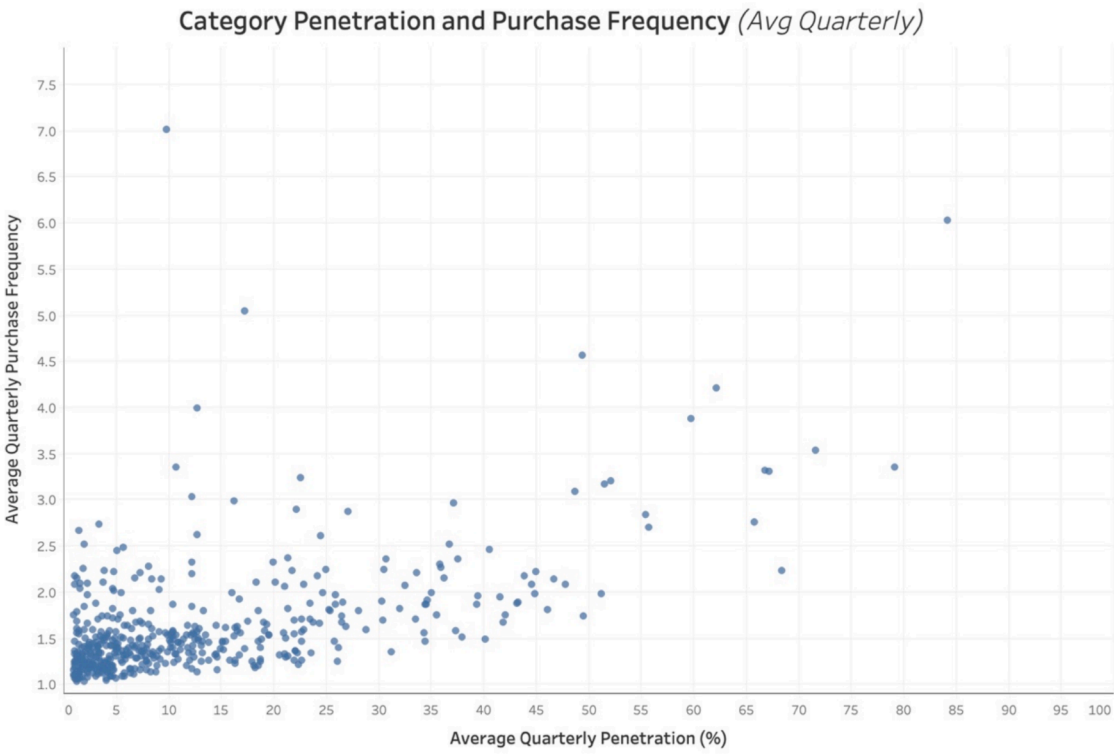


Fig. 1. Average quarterly penetration and purchase frequencies across categories (2007–2019).

Table 3
Revenue sales changes across time-periods.

Time Period	n	1-Year Revenue Sales Change (%)								
		Mean	SD	Min	Percentile (20)	Percentile (40)	Median	Percentile (60)	Percentile (80)	Max
2007–2008	474	**3.5**	8.4	−24.5	−2.0	2.0	**3.7**	5.1	8.8	45.5
2008–2009	474	**1.6**	8.1	−27.9	−4.0	0.5	**2.1**	3.4	6.7	45.3
2009–2010	474	**−3.2**	9.1	−32.6	−8.0	−4.8	**−3.7**	−2.4	1.6	119.5
2010–2011	474	**3.1**	13.4	−50.9	−2.1	1.1	**2.0**	3.3	7.4	206.1
2011–2012	474	**0.9**	10.7	−43.5	−4.9	−1.5	**0.1**	1.4	4.7	121.2
2012–2013	474	**−0.7**	9.8	−59.8	−6.3	−2.9	**−1.4**	−0.1	3.8	57.5
2013–2014	474	**0.7**	9.8	−29.4	−4.6	−1.1	**0.1**	1.7	4.8	143.3
2014–2015	474	**−2.0**	9.6	−78.6	−6.8	−3.3	**−2.1**	−0.4	3.2	80.4
2015–2016	474	**−0.3**	8.3	−38.3	−6.1	−2.0	**−0.5**	1.2	4.8	27.6
2016–2017	474	**−1.0**	13.7	−29.4	−6.8	−2.8	**−1.7**	−0.5	3.3	238.1
2017–2018	474	**−0.3**	7.6	−43.1	−5.0	−1.7	**−0.3**	1.0	4.4	30.3
2018–2019	474	**−2.2**	9.1	−76.0	−6.2	−3.6	**−2.4**	−1.1	2.0	62.1
All Years	**5,688**	**0.0**	**10.1**	**−78.6**	**−5.6**	**−2.0**	**−0.4**	**1.2**	**5.0**	**238.1**

Table 4
Revenue sales changes by category size (quarterly penetration).

Quarterly Penetration	%	1-Year Revenue Sales Change (%)								
		Mean	SD	Min	Percentile (20)	Percentile (40)	Median	Percentile (60)	Percentile (80)	Max
<5 %	37	−0.3	13.9	−78.6	−7.4	−3.0	−1.1	1.0	5.9	238.1
5–10 %	20	−0.1	8.8	−59.8	−6.1	−2.3	−0.7	1.1	5.2	80.4
10–20 %	21	0.1	6.6	−47.0	−4.5	−1.4	−0.2	1.1	4.1	47.0
20–40 %	16	0.6	5.8	−21.3	−4.0	−1.0	0.1	1.7	4.6	45.5
40 % +	7	0.5	5.2	−22.1	−3.4	−0.7	0.2	1.1	3.9	21.9
All Years	**100**	**0.0**	**10.1**	**−78.6**	**−5.6**	**−2.0**	**−0.4**	**1.2**	**5.0**	**238.1**

A correlation analysis between component changes and total revenue (Table 5) provides additional insights into these dynamics. Across the full sample, penetration changes had the strongest correlation with revenue changes ($r = 0.856$, $p < 0.01$). This relationship weakened for the highest-penetration categories (≥40 %; $r = 0.481$, $p < 0.01$), where penetration showed lesser year-to-year variability. Among these highest-penetration categories, price per volume changes had the strongest correlation with revenue changes ($r = 0.667$, $p < 0.01$). Changes in volume per trip consistently showed minor or non-significant correlations with revenue changes across all category sizes.

Table 5
Revenue sales and component changes by category size – Means, standard deviations, and correlations.

Quarterly Penetration	%	Mean Revenue Sales and Component % Changes					Standard Deviations Revenue Sales and Component % Changes					Correlations % Changes in Revenue Sales and:			
		Rev	Pen	PFreq	Vol/ Trip	Price/ Vol	Rev	Pen	PFreq	Vol/ Trip	Price/ Vol	Pen	PFreq	Vol/ Trip	Price/ Vol
<5%	37	−0.3	−2.2	−0.2	0.2	2.3	13.9	12.4	2.7	7.3	7.7	0.871**	0.275**	0.193**	0.315**
5–10 %	20	−0.1	−1.9	0.0	0.2	1.8	8.8	7.5	1.6	5.3	5.3	0.881**	0.409**	0.135**	0.162**
10–20 %	20	0.1	−1.7	−0.2	0.0	2.1	6.6	5.2	1.3	2.6	4.0	0.830**	0.530**	0.038 ns	0.354**
20–40 %	16	0.6	−1.3	−0.3	0.0	2.3	5.8	3.3	1.3	3.1	5.8	0.738**	0.634**	−0.020 ns	0.447**
40 % +	7	0.5	−0.9	−0.5	−0.2	2.2	5.2	2.0	1.4	2.5	5.4	0.481**	0.521**	−0.029 ns	0.667**
Total	**100**	**0.0**	**−1.8**	**−0.2**	**0.1**	**2.1**	**10.1**	**8.7**	**2.0**	**5.4**	**6.2**	**0.856****	**0.340****	**0.157****	**0.311****

Rev = Revenue Sales, **Pen** = Average Quarterly Penetration, **PFreq** = Average Quarterly Purchase Frequency, **Vol/Trip** = Average Quarterly Volume per Trip, **Price/Vol** = Average Quarterly Price per Volume.
** $p < 0.01$, * $p < 0.05$, ns not significant.

Category growth rates are grouped into quintiles of revenue performance; Table 6 reports the average component changes within each performance band. This grouping revealed how the four components differed across the growing, stable, and declining categories. In categories with < 20 % penetration, average penetration changes were the primary factor distinguishing the top and bottom quintile groups, highlighting penetration as the main driver of growth and decline in these categories. Different patterns emerged for high-penetration categories, especially those with ≥ 40 % quarterly penetration. Top-quintile growth in these larger categories is driven primarily by increases in price per volume, which occur without losses in penetration or purchase frequency. Conversely, penetration loss is the largest contributor to revenue decline in high-penetration categories.

In summary, addressing RQ2, penetration changes are the primary drivers of category revenue changes across categories. However, this relationship weakens in high-penetration categories, where price per volume becomes the strongest contributor to revenue growth.[2] While purchase frequency is positively correlated with revenue changes, its overall impact is limited because of its low variability. Finally, changes in volume per trip had little to no relationship with category revenue changes.

4.4. Relationships between changes in revenue sales and category growth components

To address RQ3, the correlations between the four components of category growth—penetration, purchase frequency, volume per trip, and price per volume—were analysed (Table 7). Two key relationships emerged from the analysis.

First, changes in category penetration are positively associated with changes in purchase frequency, with the strongest correlations observed in mid- to high-penetration categories. For instance, among categories with 20–40 % quarterly penetration, the correlation was $r = 0.617$ ($p < 0.01$), and similarly for categories with ≥ 40 % penetration, it was $r = 0.698$ ($p < 0.01$). Because penetration tends to exhibit greater variability than purchase frequency, larger shifts in penetration are often accompanied by smaller changes in purchase frequency.

Second, across all category sizes, a consistent negative association was observed between changes in price per volume and volume per trip ($r = -0.538$, $p < 0.01$). In contrast, the correlations between changes in price and changes in penetration ($r = 0.046$, $p < 0.01$) and purchase frequency ($r = 0.051$, $p < 0.01$) are negligible, despite achieving statistical significance because of the large dataset ($n = 5{,}688$). These results suggest that price changes have minimal impact on the number of purchase occasions but may influence the quantity purchased per occasion. The negative relationship between price and volume per trip could reflect a shift towards smaller premium-priced packs, larger value-priced packs, or stockpiling behaviour prompted by promotions.

In relation to RQ3, two main relationships were observed: a positive association between changes in penetration and purchase frequency, particularly in mid- to high-penetration categories, and a negative association between changes in price per volume and volume per trip across all categories.

5. Discussion

This study advances the knowledge of category growth by analysing patterns of revenue growth and decline across 474 CPG categories over

[2] This pattern may seem analogous to the 'triple jeopardy' effect for retail stores (Bhat & Fox, 1996; Sharp & Riebe, 2005), whereby bigger stores enjoy higher frequency and higher spend per visit. However, while growth in larger categories tends to be more associated with price rises than is the case for smaller categories, larger categories tend not to have higher average spend per purchase occasion.

Table 6
Revenue sales and component changes by category size and growth quintiles.

Categories	Quintiles (Revenue Sales Change)	Mean Revenue Sales and Component % Changes				
		Rev	**Pen**	**PFreq**	**Vol/ Trip**	**Price/ Vol**
< 5 % Quart. Pen.	Quintile 5 (Top 20 %)	15.6	9.4	0.9	0.8	5.3
	Quintile 4	3.2	0.3	0.2	0.4	2.7
	Quintile 3	−1.1	−2.3	−0.2	0.5	1.2
	Quintile 2	−5.2	−5.7	−0.5	0.0	1.3
	Quintile 1 (Bottom 20 %)	−14.2	−12.8	−1.2	−0.8	0.8
	Difference (Q5-Q1)	**29.9**	**22.2**	**2.1**	**1.6**	**4.6**
5–10 % Quart. Pen.	Quintile 5 (Top 20 %)	11.8	6.4	0.9	1.0	3.8
	Quintile 4	2.9	0.2	0.2	0.1	2.7
	Quintile 3	−0.6	−1.8	−0.2	0.1	1.4
	Quintile 2	−4.1	−4.6	−0.3	−0.1	1.2
	Quintile 1 (Bottom 20 %)	−10.6	−9.8	−0.8	0.0	0.3
	Difference (Q5-Q1)	**22.3**	**16.2**	**1.7**	**1.0**	**3.4**
10–20 % Quart. Pen.	Quintile 5 (Top 20 %)	9.1	3.6	0.7	0.1	4.6
	Quintile 4	2.5	−0.2	0.1	0.1	2.6
	Quintile 3	−0.1	−1.5	−0.1	0.1	1.6
	Quintile 2	−2.8	−3.5	−0.4	0.0	1.2
	Quintile 1 (Bottom 20 %)	−8.1	−7.1	−1.1	−0.2	0.4
	Difference (Q5-Q1)	**17.3**	**10.7**	**1.8**	**0.4**	**4.2**
20–40 % Quart. Pen.	Quintile 5 (Top 20 %)	8.7	2.2	0.8	0.1	5.6
	Quintile 4	3.1	0.1	0.2	−0.1	3.3
	Quintile 3	0.3	−1.3	−0.3	0.2	1.8
	Quintile 2	−2.5	−2.7	−0.7	0.0	1.0
	Quintile 1 (Bottom 20 %)	−6.6	−4.7	−1.5	−0.1	−0.4
	Difference (Q5-Q1)	**15.3**	**6.9**	**2.3**	**0.2**	**5.9**
40 % + Quart. Pen.	Quintile 5 (Top 20 %)	7.7	0.6	0.6	−0.2	6.9
	Quintile 4	2.5	0.2	0.0	−0.3	2.6
	Quintile 3	0.2	−0.9	−0.5	0.0	1.7
	Quintile 2	−1.8	−1.5	−1.0	−0.2	1.1
	Quintile 1 (Bottom 20 %)	−6.1	−2.7	−1.8	−0.1	−1.5
	Difference (Q5-Q1)	**13.8**	**3.3**	**2.4**	**−0.1**	**8.4**

Rev = Revenue Sales, **Pen** = Average Quarterly Penetration, **PFreq** = Average Quarterly Purchase Frequency.
Vol/Trip = Average Quarterly Volume per Trip, **Price/Vol** = Average Quarterly Price per Volume.

13 years using household panel data. The ETET approach helped identify category size (measured by quarterly penetration) as a key factor in growth dynamics. High-penetration categories demonstrated greater revenue stability, whereas categories with lower penetration exhibited larger fluctuations.

The findings highlight penetration and price per volume as the primary drivers of category revenue change. Penetration is the key growth driver in low-penetration categories, whereas price per volume becomes more influential in categories with a quarterly penetration of ≥ 40 % (equivalent to approximately 70 % annual penetration). However, penetration losses remain the leading contributor to revenue decline across all category sizes. This study also highlights the interdependencies among buyer behaviour metrics. Changes in penetration were positively correlated with purchase frequency, whereas changes in price per volume were negatively correlated with volume per trip. These interdependencies suggest that the behavioural components are not independent paths to growth but are interconnected, with shifts in one metric often aligning with changes in others.

This study formalises these insights into a category growth framework (Fig. 2), which is a significant contribution to both theory and practice. This framework has practical value for managers as it emphasises the behavioural drivers of category growth and lays the foundation for future research. Having clarified *what* occurs during category growth, subsequent research can now explore *why* and *how* these dynamics occur.

5.1. Theoretical contributions

This study advances our understanding of category growth and decline, laying the foundation for a theory grounded in buyer behaviour. Previously, category growth pathways, such as attracting more buyers, increasing purchase frequency, encouraging larger purchases, and raising prices, were largely speculative (Bang et al., 2016; Bass et al., 2005; Varadarajan et al., 1992). This study provides the first rigorous investigation into buyer behaviours underlying category revenue changes, identifying regularities in category-level buying patterns, similar to brand-level observations (Baldinger et al., 2002; Romaniuk et al., 2014, 2018). These findings suggest that the potential drivers of category growth are not as arbitrary as previously assumed (e.g., Kotler et al., 2008).

The findings revealed that strategies aimed at increasing consumption among existing buyers are generally ineffective in driving category expansion. Changes in purchase frequency are minimal compared to larger shifts in penetration, which is consistent with Dunn et al. (2021). While the volume per trip varies more, it shows little association with the overall revenue changes. Penetration and price per volume emerged as the primary drivers of category growth, providing a new perspective for evaluating marketing effectiveness. For instance, prior studies

Table 7
Correlations of changes in buyer behaviour components.

Quarterly Penetration	%	Correlations % Changes in Buyer Behaviour Components					
		Pen & PFreq	Pen & Vol/Trip	Pen & Price/Vol	PFreq & Vol/Trip	PFreq & Price/Vol	Vol/Trip & Price/Vol
<5%	37	0.191**	0.030 ns	0.101**	−0.047*	0.051*	−0.511**
5–10 %	20	0.294**	0.100**	−0.115**	−0.028 ns	0.045 ns	−0.638**
10–20 %	20	0.397**	−0.073*	−0.036 ns	0.011 ns	0.003 ns	−0.516**
20–40 %	16	0.617**	−0.007 ns	0.005 ns	−0.111**	0.132**	−0.691**
40 % +	7	0.698**	−0.018 ns	−0.089 ns	−0.066 ns	0.005 ns	−0.551**
Total	**100**	**0.239****	**0.033***	**0.046****	**−0.043****	**0.051****	**−0.538****

Pen = Average Quarterly Penetration, **PFreq** = Average Quarterly Purchase Frequency.
Vol/Trip = Average Quarterly Volume per Trip, **Price/Vol** = Average Quarterly Price per Volume.
** $p < 0.01$, * $p < 0.05$, ns not significant.

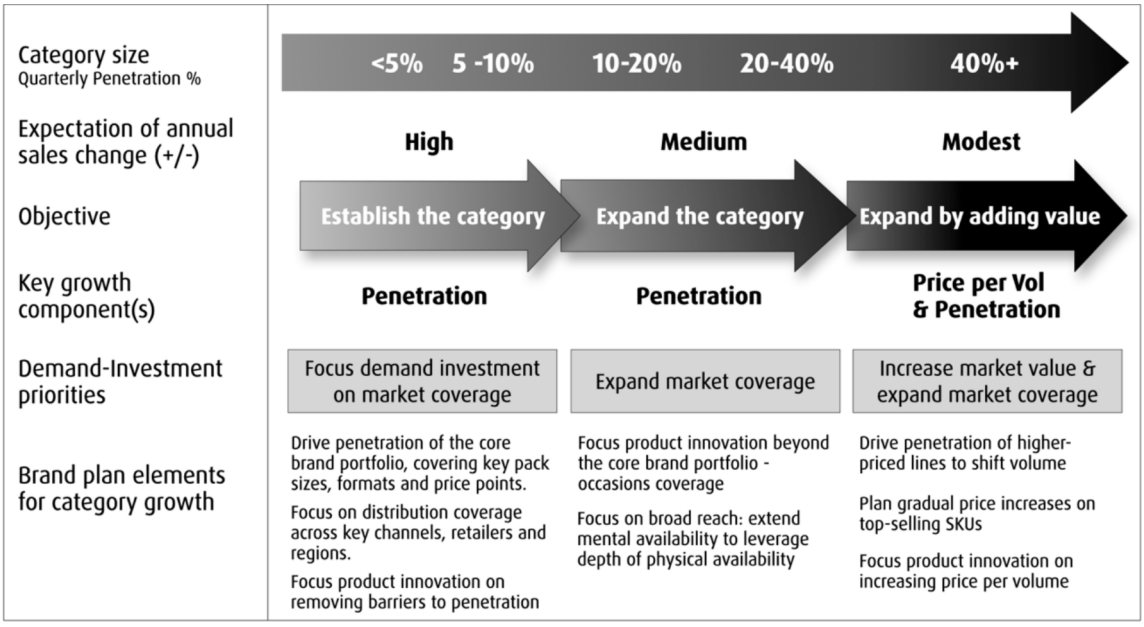


Fig. 2. Category growth framework.

observed that price promotions fail to induce long-term category expansion (Dekimpe et al., 1998; Nijs et al., 2001). When assessing price promotions in terms of growing category penetration or price per volume, their limitations are self-evident: they mainly attract existing buyers (Dawes, 2018; Ehrenberg et al., 1994) and lower the average price per volume (because they rely on price reductions).

Category size, measured as quarterly penetration, is a key factor influencing growth dynamics. Discrepancies in prior studies regarding the incidence of category sales stationarity and evolution (e.g., Dekimpe & Hanssens, 1995a; Dekimpe et al., 1998; Nijs et al., 2001; Srinivasan & Bass, 2000) may have partly resulted from analysing categories at varying penetration levels. Similarly, the stationarity observed in the category penetration by Pauwels et al. (2002) likely reflects their focus on two high-penetration categories: soup and yoghurt. A broader category sample is likely to reveal greater variation and evolution in penetration.

Differences in category size may also explain why this study and that of Tanusondjaja et al. (2022) differ in reported annual category revenue changes. Tanusondjaja et al. (2022) observed an average revenue increase of 1.1 % with lower variability (SD 3.6), which aligns closely with our findings for high-penetration categories (mean 0.5 %, SD 5.2). By contrast, our full sample of 474 categories showed no average increase or significantly higher variability (SD 10.1). This suggests that the smaller sample of 39 categories in Tanusondjaja et al. (2022) disproportionately includes high-penetration categories, potentially underestimating the variability and impact of category growth on smaller categories.

This study underscores the relevance of household panel data and buyer behaviour metrics to understand category growth. Unlike retail panel or per capita consumption data, which only reveal how much is sold, household panel data provide insights into the role of household penetration. By integrating penetration, purchase frequency, volume per trip, and price per volume into a unified framework (see Fig. 2), this study offers a foundation for exploring buyer behaviour across the different stages of category development.

5.2. *Managerial implications*

Category growth has become a central focus for brand strategists in established product markets, as highlighted in the Introduction, with references to statements from P&G, Mars Wrigley, and PepsiCo Europe. This study highlights several practical implications for understanding and developing effective category-first strategies. Identifying penetration and price per volume as consistent growth drivers provides a clear strategic direction for category expansion. Conversely, strategies centred solely on increasing consumption among existing buyers (e.g. purchase frequency or volume per trip) should be de-prioritised, as they are unlikely to succeed and could result in costly mistakes.

The findings also underscore the importance of recognising category size, measured by quarterly penetration, as a critical factor influencing growth dynamics. This understanding enables marketers to set realistic growth ambitions and align strategies accordingly. Low-penetration categories tend to experience greater sales fluctuations driven by changes in penetration, whereas high-penetration categories exhibit more stable sales. Such insights allow strategists to assess the risks of investing in categories of varying sizes, similar to evaluating risk profiles in financial markets.

Category size also dictates the most effective growth strategy. For lower penetration categories, the focus should be on expanding penetration, which is the primary driver of growth. In high-penetration categories, strategies that increase price per volume, such as selective price increases or premiumisation (Dawes et al., 2019), become more relevant, provided that significant sales volume losses are avoided. Regardless of category size, maintaining penetration is essential to prevent revenue decline.

Adopting a category-first approach has implications for portfolio prioritisation, new product innovation, and brand marketing activities, as Fig. 2 illustrates. Understanding the implications of a category's penetration level allows managers to audit portfolios and innovation pipelines to support category growth.

In some categories, low penetration reflects that many households have yet to make a purchase, while in others, a large buyer base exists, but only a small fraction is captured within a quarter or year because of infrequent purchasing (e.g. shoe polish or fly spray). This distinction is important for determining appropriate strategies. For categories in which low penetration results from few prior purchases, the focus should be on expanding market coverage across key channels, retailers, pack sizes, and formats to match core usage occasions. Opening price points and trial packs can play a critical role in encouraging trial among new buyers. Innovation in the early maturity stages should focus on breaking category penetration barriers, which are usually associated with a pack for a particular channel (e.g. a pack for Aldi or Costco), a format to address a key usage occasion, or a variant to accommodate a buyer cohort (e.g. gluten-free, zero sugar).

As the category matures, growth can be maintained by expanding into usage occasions beyond the core, and investing in mental availability. However, as growth slows, and the category reaches a broader base, the portfolio must include options that support higher price points. Examples of such innovations include ready-to-heat rice, coffee capsules, and running shoes with carbon-fibre plates. A critical consideration is whether such premium products already exist in the portfolio and require additional investment or whether a gap needs to be addressed through new introductions. By overlooking this question, marketers may risk under-developing the typically small premium segment of the portfolio.

Overall, this framework helps marketers in global businesses avoid the pitfalls of selecting innovations from a generic 'menu', which can unnecessarily complicate the portfolio and even hinder category expansion in specific markets. For example, Gatorade faces different portfolio requirements for category growth in a market where sports drinks have 5 % quarterly penetration compared to those with 35 % quarterly penetration.

6. Future research agenda

Category growth remains underexplored in academic research despite its emerging importance for industry decision-makers. A research agenda for developing and building upon this framework is proposed.

6.1. Empirical generalisations for category growth

This study identified empirical regularities in the annual growth and decline of US CPG categories, providing a foundation for developing empirical generalisations, highlighting relationships repeatedly observed across various conditions (Uncles & Kwok, 2013). Strengthening these generalisations requires replication and extension to test generalisability and uncover boundary conditions (Ehrenberg, 1995; Uncles & Kwok, 2013). Future studies should investigate different markets, particularly in emerging economies with different retail structures, disposable income, and category development. Studies can examine dynamic contexts, such as those shaped by continuous innovation (e.g. tech sectors) or where categories are merging, splintering, or emerging. Insights from such studies would refine our understanding of how categories evolve, supporting strategies for managing growth and decline.

6.2. Extending theory through models of brand buying and brand competition

Establishing links between new empirical regularities and existing theories is a productive path for theory development (Barwise, 1995). Category penetration and purchase frequency are inputs for the NBD and NBD-Dirichlet models, which underpin long-established theories of repeat-buying behaviour (Ehrenberg et al., 2004; Ehrenberg, 1988; Goodhardt et al., 1984; Mizerski et al., 2013; Sharp et al., 2012; Trinh & Lam, 2016; Uncles & Kwok, 2009). These models assume stationarity and can be applied as benchmarks to evaluate the observed changes in category buying, providing a basis for understanding how category growth extends NBD-Dirichlet theory. A novel approach would involve examining how the parameters of the NBD-Dirichlet model (Driesener & Rungie, 2022) change over time as category sales grow and decline (see for example Sharp et al., 2012).

This study identified a positive relationship between category penetration and purchase frequency, both cross-sectionally and over time. This pattern suggests that larger categories tend to attract more buyers who, on average, also purchase from the category more frequently. While this parallels the well-documented double jeopardy effect at the brand level—where larger brands attract buyers who buy more often (Ehrenberg et al., 1990; Graham et al., 2017; Greenacre et al., 2015; Habel & Lockshin, 2013)—this relationship at the category level warrants further investigation. Understanding this pattern at the category level can provide insights into how and where categories compete for sales.

An important distinction at the category level is the role of substitutes and complements. Unlike brand substitution within a category, which underpins the NBD-Dirichlet model's predictions (Sharp et al., 2024), category-level dynamics can involve varying degrees of substitutability and complementarity. For instance, in some cases, category growth may come at the expense of competing categories, whereas in others, categories may grow together because of complementary consumption patterns (e.g. pasta and pasta sauce).

A more comprehensive understanding of category competition can also be developed by extending the theories of mental availability (Romaniuk, 2013, 2023) to the category level. Modelling category-level mental availability metrics would shed light on how and when categories share buyers, offering insights into competition across both mental and physical availability (Sharp et al., 2024).

6.3. How categories grow price per volume

This study identified price per volume as a key driver of revenue growth, especially in high-penetration categories. Future studies should examine how price per volume increases can occur without reducing purchase occasions or even while expanding them. It is essential to distinguish between simple price increases and premiumisation. Price increases generally reduce sales volumes (Andreyeva et al., 2010), whereas premiumisation involves a shift to higher-priced products (Dawes et al., 2019). Research should investigate the conditions under which each strategy effectively drives category growth through price per

volume and identify the key drivers of premiumisation, including premium brands, changes in price-pack architecture, and product innovations.

This study examined category growth and decline from 2007 to 2019, a period characterised by low inflation. However, in recent years, inflationary pressures have introduced significant variability in their effects across categories (Dekimpe & van Heerde, 2023). The findings provide benchmarks for assessing category performance during periods of economic volatility and a framework for investigating how higher prices affect buyer behaviour.

6.4. Marketing and non-marketing effects on category growth

While this study takes a descriptive approach to how categories grow, future research should explore the causes of changes in buyer behaviour. It is important to determine whether these patterns are driven by marketing activities or external factors such as macroeconomic conditions and consumer trends (e.g. health consciousness and sustainability). Understanding these external drivers will help marketers decide whether to drive category growth or to adapt to broader market trends. Further research on how marketing tactics influence buyer behaviour is necessary. Previous studies have focused on price promotions and advertising, but incorporating buyer behaviour and category size could provide new insights. Research should also examine the effects of distribution, portfolio management, and product innovation.

Finally, this study focused on demand-side behaviour. Expanding research to include supply-side factors, such as new category emergence, industry costs, and margins, would provide a more holistic understanding of category dynamics and profitability. Combining demand- and supply-side perspectives is essential for a comprehensive view of category performance and strategic decision-making.

CRediT authorship contribution statement

Steven Dunn: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Magda Nenycz-Thiel:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Charles Graham:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **John Dawes:** Writing – review & editing, Methodology, Conceptualization. **Nick Danenberg:** Writing – review & editing, Methodology, Conceptualization. **Arry Tanusondjaja:** Writing – review & editing, Methodology, Conceptualization. **Giang Trinh:** Writing – review & editing, Methodology, Conceptualization. **Bruce McColl:** Writing – review & editing, Methodology, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in order to assist with editing and improving the readability of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. ChatGPT was not used to process, analyse, or interpret any data.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Researcher(s)' own analyses calculated (or derived) based in part on data from Nielsen Consumer LLC and marketing databases provided through the NielsenIQ Datasets at the Kilts Center for Marketing Data Center at The University of Chicago Booth School of Business. The conclusions drawn from the NielsenIQ data are those of the researcher(s) and do not reflect the views of NielsenIQ. NielsenIQ is not responsible for, had no role in, and was not involved in analyzing and preparing the results reported herein.

Data availability

The dataset used in this paper can be accessed through the Chicago Booth Kilts Centre for Marketing (https://www.chicagobooth.edu/research/kilts/research-data/nielseniq)

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Steven Dunn is a Research Fellow at the Ehrenberg-Bass Institute for Marketing Science, University of South Australia. His key areas of interest include understanding how categories grow and decline, the effects of pricing and price promotions, and long-run patterns in buying behaviour. He has published in journals such as Journal of Marketing Management, Journal of Strategic Marketing, Journal of Service Management, Journal of Retailing and Consumer Services, and International Journal of Market Research.

Magda Nenycz-Thiel is a Research Professor at the Ehrenberg-Bass Institute. Magda was the first Mars-funded chair - a position fully sponsored by Mars Incorporated. Magda, together with a research team, pioneered the category growth research at the Institute and has been focusing on applying the discoveries to improve how growth investments are made. Magda's main areas of research are category growth, physical availability, private labels, portfolio management, e-commerce and innovation. She has published widely in International journals from the Journal of Advertising Research, and Marketing Letters to the European Marketing Journal.

Charles Graham is a visiting professor at the Ehrenberg-Bass Institute and at the London South Bank University Business School. Prior to becoming an academic he held board-level roles with responsibilities for consumer packaged-goods and hospitality brands in the UK and Middle East. His research spans many facets of marketing practice, including behavioural drivers of long-run consumer brand growth; high street vitality and place marketing effectiveness; advertising and promotion response; and empirical generalizations in buying through B2B supply chains. He has published in leading marketing journals and presents regularly at international conferences and to strategic marketers and their agencies.

John Dawes is a Professor of Marketing and an Associate Director of the Ehrenberg-Bass Institute. John's research interests center around buyer behavior, loyalty/retention, and pricing. He has also conducted research on customer satisfaction and word of mouth, and their possible links to brand growth. John has published in leading journals such as Journal of Service Research, Journal of Retailing, Journal of Business Research, European Journal of Marketing, Marketing Letters among others.

Nick Danenberg is a founding researcher at the Ehrenberg-Bass Institute for Marketing Science. He is committed to producing and disseminating scientific knowledge about marketing to help brands grow. His research focuses on advertising and media effectiveness, consumer behaviour and targeting. Most recently he has focused on developing new methods of fitting the NBD-Dirichlet model through novel ways of deriving the model parameter estimates.

Arry Tanusondjaja is a Senior Marketing Scientist at the Ehrenberg-Bass Institute. Prior to academia, he held various management roles in marketing analytics, business analysis, and market intelligence in multiple South East Asian countries and Australia. His research interests span across product portfolio management, consumer behaviour, category growth, media scheduling, and market competition. He has published in various journals that include the Journal of the Academy of Marketing Science, Journal of Strategic Marketing, International Journal of Market Research, Journal of Retailing and Consumer Services, and Marketing Letters.

Giang Trinh is an Associate Professor at UniSA Business and Senior Marketing Scientist at the Ehrenberg-Bass Institute. His current research focuses on quantitative method development and applications in the areas of consumer purchasing behavior, brand competition, market structure, price promotion, distribution, advertising and sales relationship, and new product launch. His work can be found in Journal of Business Research, Marketing Letters, European Journal of Marketing, Journal of Marketing Management, Journal of Retailing and Consumer Services, International Business Review, International Journal of Market Research, Journal of Product and Brand Management, Journal of Consumer Behavior and Australasian Journal of Marketing.

Bruce McColl retired from Mars in 2016 after 10 years as Global Chief Marketing Officer, and total of 24 years with the company. He is regarded a champion of both creativity and science. Following Mars, Bruce joined the Ehrenberg-Bass Institute as Industry Professor, where his unique experience in the application of scientific knowledge to marketing practice benefited the Institute's global industry partners and helped drive new research. Bruce retired from the Institute in October, 2022.